Yuhyun Lee

In vivo & Ex vivo rodent experiments | Intracortical recording | Signal processing

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PROFESSIONAL PROFILE

Biomedical Engineering Ph.D. candidate at **Carnegie Mellon University** specializing in **neuromodulation, neural interfaces**, and **preclinical validation**, with **hands-on experience** with in vivo and ex vivo **electrophysiology, electrical brain stimulation**, and **experimental system development**.

- **Research Focus**:
  - Hands-on ex vivo and in vivo experiments in rodents, as well as in vivo experiments in zebrafish.
  - High-density electrophysiological recordings during minimally invasive electrical brain stimulation and under pharmacological conditions.
  - End-to-end development of novel animal models with hardware and software integration for system automation.
  - FEM simulations, signal processing, signal QC, statistical and quantitative analyses of stimulation-evoked neural responses, applied machine learning, histology, image reconstruction, and spike detection for in-depth analysis of experimental data.
- **Lab-to-Market:** U.S., EU, and Korean patents; zebrafish EEG technology licensed to an industry partner through technology transfer.
- **Publications:** Biosensors and Bioelectronics, Sensors, Journal of Neural Engineering, etc.
- **Honors:** CMU Presidential Fellowship, Dowd Fellowship, IEEE Next Gen Scholar

CORE SKILLS

- **Preclinical experiments**: Stereotaxic mouse procedures, Electrophysiology (Multiunit activity, fiber volley recordings, LFP, EEG, brain slice recordings, extracellular current-clamp recordings), Brain stimulation (Electrical, chemical), 64-channel recording, PFA tissue fixation, fluorescence microscopy
- **Electrophysiology devices:** Intan RHD, Ripple Neuro, Digitimer DS8R/DS5, Multiclamp 700B, Axon Digidata 1550B, BIOPAC
- **Data and modeling:** MATLAB (signal processing, statistical analysis, time-series analysis, GUI development, visualization, and workflow automation); applied ML (RBF-SVM and cross-validation); COMSOL (electric-field, thermal, and fluid modeling); COMSOL LiveLink for MATLAB
- **Engineering:** PCB/FPCB design; KiCad; SolidWorks; CAD; 3D printing; PDMS-based microfluidic channel fabrication; parylene-C-based electrode fabrication, FIJI, ImageJ

RESEARCH EXPERIENCE

I. **Doctoral Research** | Carnegie Mellon University | 2021–Present

1) In vivo transnasal electrical stimulation (TnES) platform development

- Designed and conducted in vivo mouse TnES experiments using a 32-site transnasal electrode array and 64-channel stereotaxic recordings spanning the cortex, hippocampus, thalamus, and hypothalamus.
- Benchtop-tested the stimulation system.
- Automated stimulation-electrode selection to streamline 132 stimulation conditions (up to 2,640 trials per animal) without manual rewiring.
- Developed a two-stage spatial interpolation method to estimate the transnasal electrode positions that could not be directly visualized.
- Tested various stimulation conditions across separate current-amplitude, pulse-width, polarity, and electrode-pair protocols to characterize responses.
- Conducted postmortem controls to distinguish neural activity from stimulation artifact.
- Developed a MATLAB analysis pipeline for biosignals to quantify regional and temporal neural response profiles by applying stimulation artifact removal, bandpass filtering, and spatial filtering.
- Performed fluorescence microscopy for probe tracing and c-Fos staining following intracardiac perfusion with PFA.
- Developed histology software that registers serial brain sections to a brain atlas and reconstructs probe trajectories.

2) Modeled neural response thresholds as a function of electric-field orientation in ex vivo brain slices

- Conducted ex vivo brain-slice stimulation using orthogonal electrode arrays to test how electric-field orientation affects neural activation.
- Used COMSOL LiveLink for MATLAB to run FEM electric-field simulations and extract electric-field features; performed mesh convergence and thermal-safety analyses in COMSOL Multiphysics.
- Performed statistical analyses of neural responses under different stimulation conditions.
- Integrated 538 observations from 21 mouse brain slices with FEM-derived E-field features and applied a class-balanced RBF-SVM with fivefold cross-validation, estimating >5× variation in activation threshold across electric-field orientations.
- Applied the classifier developed from ex vivo experimental data to electric fields simulated using a human head model (ROAST).

3) Collaborations

- Set up human transcranial electrical stimulation (TES) to measure EMG.
- Measured electric fields from a human cadaver head.
- Hormone assays using ELISA.

II. **Master's Research |** DGIST, Daegu, Korea | 2018–2020

Thesis: Development of EEG recording systems from multiple adult and larval zebrafish

- Led end-to-end development and validation of parallel zebrafish EEG platforms, from Faraday cage and recording-system setup through animal experiments, signal analysis, and pharmacological validation.
- Wrote IACUC approval documents.
- Developed a 3D-printed platform for simultaneous EEG recording from four adult zebrafish and a microfluidic platform for four larvae; used COMSOL fluid-flow simulations to support continuous drug delivery, achieving 75–78% recording success and reducing adult-fish recording time and cost fourfold.
- Analyzed multichannel EEG spectra, ictal events, spiking and bursting activity, quantifying antiepileptic drug efficacy through a 60% reduction in convulsant-induced amplitude increases in adults and a 72% reduction in ictal events in larvae.
- Co-invented zebrafish EEG technology that was patented in Korea and the EU and licensed to an industry partner for commercialization.

PATENTS & TECHNOLOGY TRANSFER

- M. Forssell, Y. Guo, **Y. Lee**, V. Jain, P. Grover, Sinus Implant for Deep Brain Stimulation and Sensing, International Patent Application Publication No. WO 2026/072884, Apr. 2026.
- M. Forssell, Y. Guo, **Y. Lee**, et al. "Method for Non-Invasive or Minimally-Invasive Stimulation of Deep Brain Targets", U.S. Patent Application Publication No. US 2024/0416125 A1, published Dec. 19, 2024.
- S. Kim, **Y. Lee**. Fish Bio Signal Measuring Device. Korean Patent No. 10-2237282 and EU Patent No. 3725219. *Licensed to an industry partner for commercialization through DGIST technology transfer.

SELECTED PUBLICATIONS

- **Y. Lee**, V. Jain, M. Chamanzar, M. Forssell, P. Grover, "Transnasal approach as a minimally invasive electrical brain stimulation targeting the arcuate nucleus in the hypothalamus in an in vivo mouse setup" *(In Preparation)*
- **Y. Lee***, Y. Guo*, M. Forssell, P. Grover, "Incorporating electric-field orientation into human-scale modeling of transnasal electrical stimulation," (*equal contribution) *(In Preparation)*
- M. Forssell, Y. Guo, **Y. Lee**, et al., "Transnasal Minimally Invasive Deep Brain Sensing and Stimulation for Mental Disorders", *Nature Communications Medicine*. *(Accepted)*
- **Y. Lee**, V. Jain, Y. Guo, M. Chamanzar, P. Grover, M. Forssell, "Influence of out-of-plane electric fields in an ex vivo model of transcranial electrical stimulation," *Journal of Neural Engineering,* Oct. 2026. DOI: 10.1088/1741-2552/aeb04a
- Y. Guo, M. Forssell, D. M. Kusyk, V. Jain, I. Swink, O. Corcoran, **Y. Lee**, et al., "DeepFocus: a transnasal approach for optimized deep brain stimulation of reward circuit nodes," *Journal of Neural Engineering*, 22(1), 016048, Feb. 2025. DOI: 10.1088/1741-2552/adac0c
- **Y. Lee**, H.W. Seo, K.J. Lee, J-W. Jang and S. Kim, "A Microfluidic System for Stable and Continuous EEG Monitoring from Multiple Larval Zebrafish", *Sensors*, *20*, 5903, Oct. 2020. DOI: https://doi.org/10.3390/s20205903
- **Y. Lee**, K.J. Lee, J-W. Jang, S. Lee, and S. Kim, "An EEG system to detect brain signals from multiple adult zebrafish", *Biosensors and Bioelectronics*, vol. 164, 112315, Sep. 2020. DOI: 10.1016/j.bios.2020.112315

CONFERENCES (FIRST AUTHOR ONLY)

- **Y. Lee**, V. Jain, P. Grover, M. Forssell, "Transnasal electrical stimulation (TnES) of the deep brain in an in vivo mouse model", NeuroBoston, Boston, MA, USA, Dec. 2, 2026.
- **Y. Lee**, V. Jain, P. Grover, M. Forssell, "Transnasal electrical stimulation (TnES) of the deep brain in an in vivo mouse model", Society for Neuroscience, Washington, D.C., USA, Nov. 14–18, 2026.
- **Y. Lee**, V. Jain, P. Grover, M. Forssell, "Transnasal electrical stimulation (TnES) of the deep brain in an in vivo mouse model", IEEE Brain Discovery and Neurotechnology Workshop, Washington, D.C., USA, Nov. 11–13, 2026. (Travel support)
- **Y. Lee**, V. Jain, P. Grover, M. Forssell, "Transnasal electrical stimulation (TnES) for brain modulation in an in vivo mouse model", 2026 NYC Neuromodulation, New York City, NY, USA, Jul. 30–Aug. 3, 2026.
- **Y. Lee**, Y. Guo, V. Jain, M. Chamanzar, P. Grover, and M. Forssell, "Predicting neural responses to *DeepFocus*, a new technique for transnasal, deep-brain electrical stimulation", Society for Neuroscience, Chicago, IL, USA, Oct. 5–9, 2024.
- **Y. Lee**, Y. Guo, V. Jain, D. Kusyk, et al., "*DeepFocus*: A new transnasal approach for minimally-invasive deep-brain stimulation: Simulation, cadaver, and ex vivo studies to predict responses in the human head", Carnegie Mellon Forum of Biomedical Engineering, Virtual, Sep. 20, 2024
- **Y. Lee**, V. Jain, M. Chamanzar, P. Grover, and M. Forssell, "Ex vivo studies of efficacy of DeepFocus: a technique for minimally invasive deep-brain stimulation," In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pp. 1-7, Jul. 2024. DOI: 10.1109/EMBC53108.2024.10781751 (Travel support)
- **Y. Lee**, V. Jain, P. Grover and M. Forssell, "An ex vivo model for non-invasive brain stimulation utilizing perpendicular planes of electrodes", Society for Neuroscience, Washington, D.C., USA, Nov. 11-15, 2023.
- **Y. Lee** and S. Kim, "A system to record EEG from multiple adult zebrafish," *IEEE EMBC*, Berlin, Germany, Jul. 23–27, 2019.
- **Y. Lee** and S. Kim, "Recording EEG signals from epilepsy-induced zebrafish treated with anti-epileptic drugs," *International Zebrafish Conference*, Suzhou, China, Jun.12–16, 2019.
- **Y. Lee** and S. Kim, "Liquid exchangeable microfluidic channels to align zebrafish," *The Korean MEMS Conference (KMEMS)*, Jeju, Korea, Apr. 4-6, 2019

SELECTED AWARDS

- Philip and Marsha Dowd Engineering Seed Fund for Graduate Student Fellowships | Pittsburgh, PA, 2024
- IEEE EMBC NextGen Scholar Award, IEEE Annual International Conference of EMBS | Orlando, FL, 2024
- Bradford and Diane Smith Graduate Fellowship in Engineering, College of Engineering, CMU | Pittsburgh, PA, 2024

EDUCATION

Ph.D. Candidate, Biomedical Engineering | Carnegie Mellon University, Pittsburgh, PA | *Advisor: Pulkit Grover* | 2021–Present

M.E., Robotics Engineering | DGIST, Daegu, Korea | Advisor: Sohee Kim | Feb. 2020*;*

B.E., Engineering | DGIST, Daegu, Korea | Feb. 2018

COURSES

Neural Engineering & Computational Methods

Neural Signal Processing, Neural Data Analysis, Fundamentals of Computational Biomedical Engineering, Signals and Systems, Introduction to BMI, Measurement and Signal Processing in Medicine, Biomedical Engineering

Biology

Neuroscience, Cell Biology, Genetics, Anatomy, Physiology

Engineering

Basic Mechanics and its Mathematical Analysis, Fluid Mechanics, Dynamics, Introduction to Control Engineering, Robots for Humans, Electric Circuits and Measurement

Mathematics

Biostatistics, Probability and Statistics, Linear Algebra, Integrated Engineering Mathematics

Material engineering

Device Fabrication Process for Nanotechnology, Introduction to Organic/Inorganic Chemistry

Enterprise

Science and Technology Policy, Technology Ventures, Intellectual Property Rights